

B.Tech. Degree VI Semester Special Supplementary Examination November 2012

ME 602 DYNAMICS OF MACHINERY (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer ALL questions)

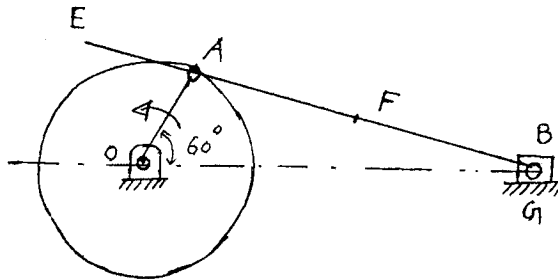
(8 × 5 = 40)

- I. (a) Define:
- (i) Kinematic pair
 - (ii) Kinematic chain
- (b) State and explain D'Alembert's principle.
- (c) Derive the expression used for the gyroscopic couple.
- (d) Explain the terms sensitiveness, hunting and stability relating to governors.
- (e) How the different masses rotating in different planes are balanced?
- (f) What is swaying couple? Derive an expression for the same.
- (g) Derive the relation for ratio of belt tensions in a flat-belt drive.
- (h) With the help of a neat sketch explain the working of a block or shoe brake.

PART B

(4 × 15 = 60)

- II. In a slider-crank mechanism the crank is 480mm long and rotates at 20 rad/s in the counter-clockwise direction. The length of the connecting rod is 1.6m. When the crank turns 60° from the inner-dead center, determine: (i) velocity of the slider (ii) velocity of a point E located at a distance 450mm on the connecting rod extended. (iii) position and velocity of a point F on the connecting rod having the least absolute velocity (iv) angular velocity of the connecting rod (v) velocity of rubbing at the pins of the crank shaft, crank and the cross head having diameters 80mm, 60mm and 100mm respectively.



OR

- III. The connecting rod of a vertical high-speed engine is 600mm long between centers and has a mass of 3Kg. Its center of mass lies at 200mm from the big end bearing. When suspended as a pendulum from the gudgeon pin axis, it makes 45 complete oscillations in 30 seconds. The piston stroke is 250mm. The mass of the reciprocating parts is 1.2 Kg. Determine the inertia torque on the crankshaft when the crank makes an angle of 140° with top-dead center. The engine speed is 1500 rpm.

(P.T.O.)

- IV. A machine shaft running at 200rpm requires a torque which varies uniformly from 1200Nm to 3600 Nm during the first half revolution, remains constant for the next one revolution, decreases uniformly to 1200Nm during the next one revolution and then remains constant for the next two revolutions, thus completing a cycle of operations. Motor has a constant torque and has a rotor of mass 50Kg with radius of gyration of 0.25m. If, in addition a flywheel of mass 2000Kg and radius of gyration 0.60 is fitted to the shaft, find (i) the power of the motor (ii) percentage fluctuation of speed.

OR

- V. In a governor of the Hartnell type, the mass of each ball is 1.5Kg and the lengths of the vertical and horizontal arms of the bell crank lever are 100mm and 50mm respectively. The fulcrum of the bell crank lever is at a distance of 90mm from the axis of rotation. The maximum and minimum radii of rotation of balls are 120mm and 80mm and the corresponding equilibrium speeds are 325 and 300rpm. Find the stiffness of the spring and the equilibrium speed when the radius of rotation is 100mm.
- VI. Four masses A, B, C and D carried by a rotating shaft at radii 80mm, 100mm, 160mm and 120mm respectively are completely balanced. Masses B, C and D are 8 Kg, 4Kg and 3Kg respectively. Determine the mass A and the relative angular positions of the four masses if the planes are spaced 500mm apart.

OR

- VII. A four cylinder inline marine oil engine has cranks at angular displacement of 90° . The outer cranks are 3m apart and inner cranks are 1.2m apart. The inner cranks are placed symmetrically between the outer cranks. The length of each crank is 450mm. If the engine runs at 90 rpm and the mass of reciprocating parts for each cylinder is 900Kg, find the firing order of the cylinders for the best primary balancing force of reciprocating masses. Also determine the maximum unbalanced primary couple for the best arrangement.

- VIII. An open-belt drive is required to transmit 10KW of power from a motor running at 600rpm. Diameter of the driving pulley is 250mm. The speed of the driven pulley is 220 rpm. The belt is 12mm thick and has a mass density of 0.001 g/mm^3 . Safe stress in the belt is not to exceed 2.5N/mm^2 . The two shafts are 1.25m apart. The coefficient of friction is 0.25. Determine the width of the belt.

OR

- IX. A band brake of drum with diameter 400mm provides a braking torque of 2KN-m. One end of the band is attached to a pin 100mm from the fulcrum of the lever and the other end to another pin 20mm from the fulcrum and on the other side of it where operating force is also acting. The angle of lap on the drum is 225° and the coefficient of friction is 0.3. Find the operating force required at the end of the lever 600mm long for the two directions of rotation of the drum.